Internet Protocol (IP)

- IP
  - Designed to connect networks
  - Work at Layer 3
    - So it Routes (what does it do in one word) → picks path across the network
  - IP Packet transmissions — arrives on layer 2, passed to layer 3
    - Routing Table
    - IP Packet Transmissions
      - connectionless
      - unreliable → throws it away if router congested
        - figures TCP will recover it.

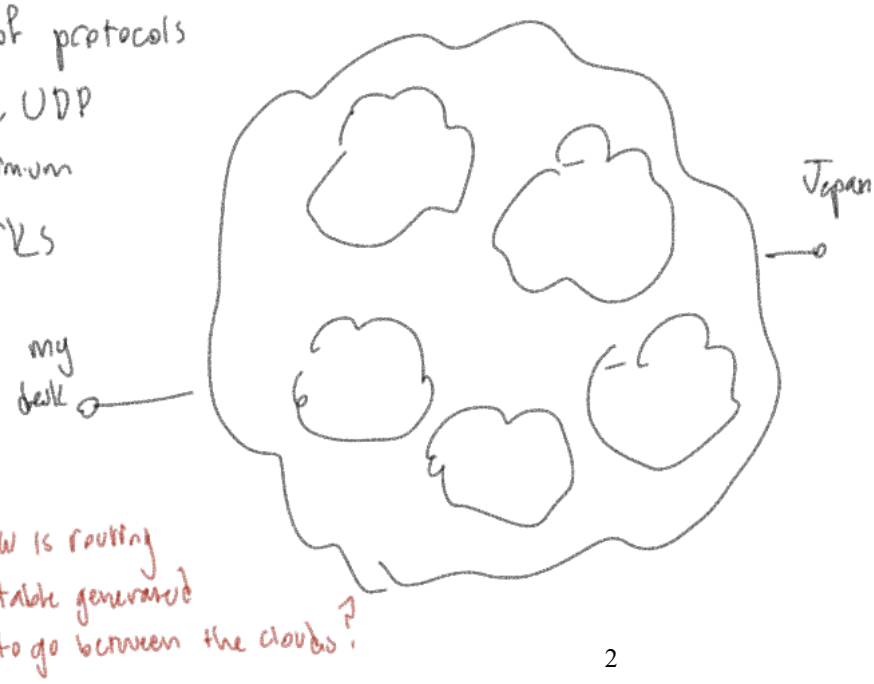
does packet switching: every packet has a header with a destination address

when packet arrives, routing table is searched for the next hop.

What is the Internet?

- You use it every day…. So what is it?

1. Runs the IP family of protocols
   - IP, TCP, UDP
     - IP: bare minimum
2. Collection of Networks



Most everyone runs pure IP

How is routing table generated to go between the clouds?

IP Routing

- Calculate routing table
  - **Router actively learns** about the network – routing protocol
  - Creates:

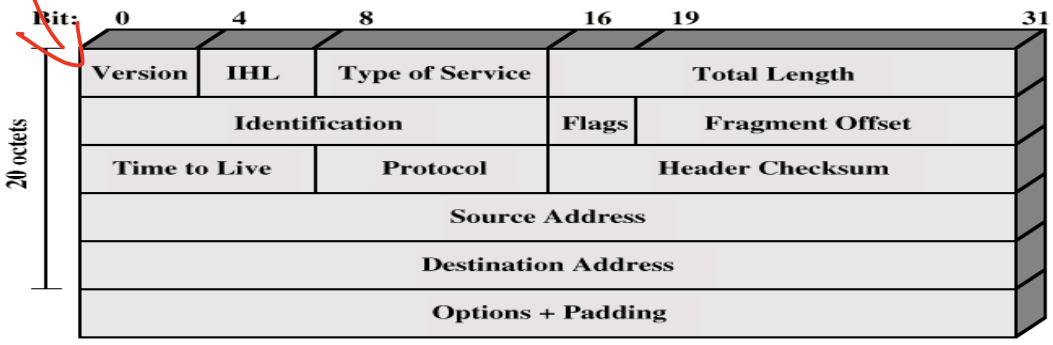
- Packet Forwarding (uses Packet Switching)

  - SEE SLIDE 1
  -
  -

The IP Header

- IP Header
  - 20 Byte fixed part, variable length option field is 40 bytes

  - Version = IPv4 or IPv6
  - IHL = Header Length in words (x4 to get bytes) → IPv6 fixed in size
  - Type of Service (ToS)
    - Also - first 6 bits called DiffServ
    - Last 2 – Explicit congestion notification bits
  - Total Length (IP PDU Length)= Length of Header + Data



Quality of Service (QoS)

give priority to some traffic over other traffic

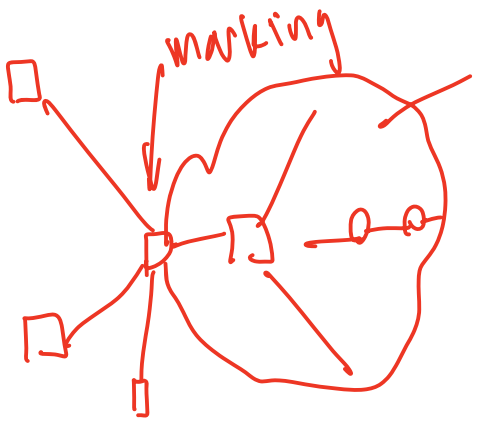
- ToS – Type of Service Bits (8 Bits)
  - First 3-bits precedence bits (deprecated/never used)
  - First 6-bit Differentiated Services bits (DiffServ)
    - These bits called: Differentiated Services Codepoint (DSCP)

64 classes of traffic

  - Remaining 2 bits are used by a TCP flow control protocol (later in the quarter)

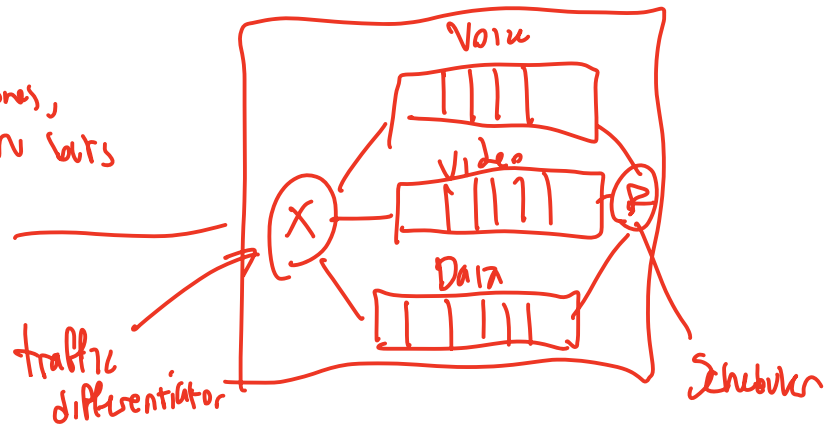
- Router Queuing and QoS
  - Edge Router – Traffic Marking
  - Per-Hop Behavior and packet forwarding (Queueing in the routers)

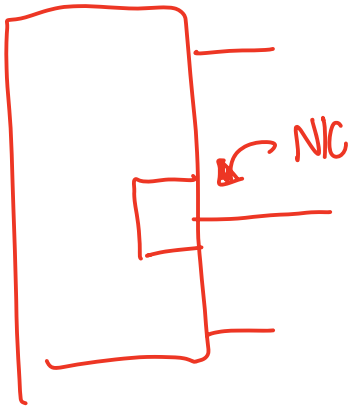
Router Queueing and QoS



within this implements DIFFSERV Queuing

you say that these are phones, you set your diffserv bits





- based on DIFFSERV bits
- determines what queue it goes into
  - physical memory

The IP Header (con't)

- IP Header
  - Fragmentation
    - Identification = Used for fragmentation and reassembly
    - Flags = (DF , don't fragment, MF, more fragments)
    - Fragment Offset = byte offset into the packet
  - Time to Live = TTL
  - Protocol = UDP, TCP

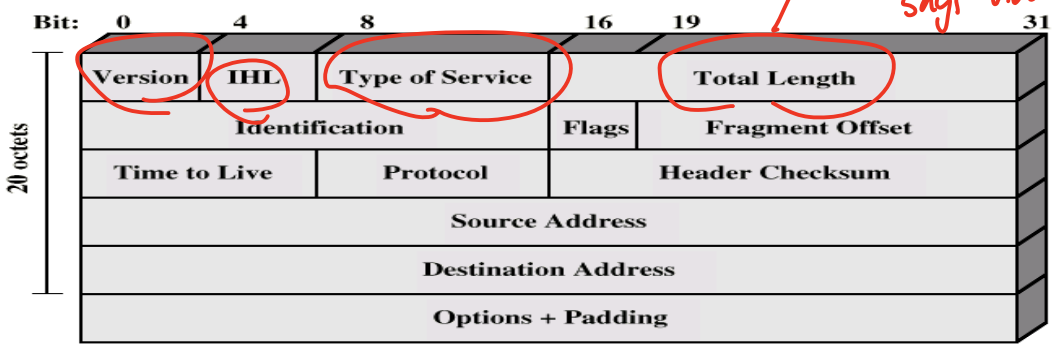
splits data apart

put back together

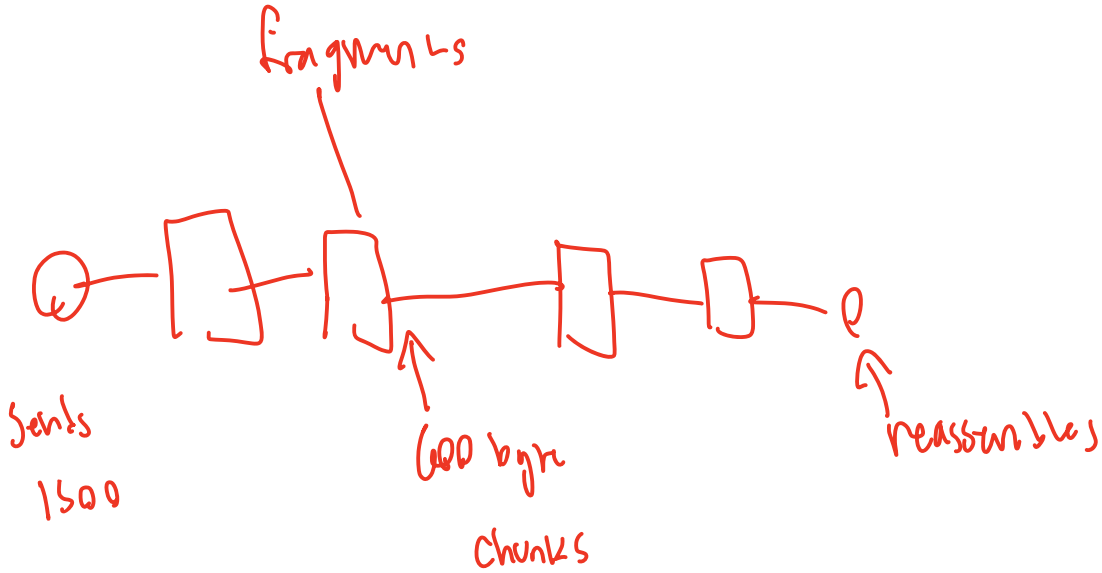
- fragmented on any router
- pieced together by receiver

n → h

says how big payload is



Fragmentation and Reassembly



Internet Protocol (IP) Addressing – IPv4

Layer 3: Routing

How to address, how does it learn about that

- **IPv4** Address Properties
  - IPv4 address is a 32 bit number (~4 billion unique addresses) MAX
    2^{32}
  - Written in Dot Decimal notation: 129.65.44.7
    255 255 255 255 val
    8 8 8 8 bits
    byte_byte_byte_byte
  - Each 8 bit number is called an Octet → byte
  - Address is hierarchical
    - Global to local
    - Think Zip code to street address

    129.65 | 44.7

    gets it to your desk

    gets exactly where its going using zipcode

Where do IP's come from?

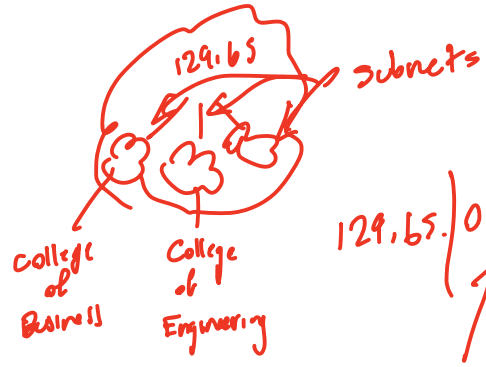
- Internet Assigned Numbers Authority IANA
  - Global IP address control
  - Regional Internet Registry (RIR)
    - 5 of these
    - ARIN – where our Internet Service Providers (ISP) get their IP addresses → American Regional Internet Registry

Registry	Area Covered
AFRINIC	Africa Region
APNIC	Asia/Pacific Region
ARIN	Canada, USA, and some Caribbean Islands
LACNIC	Latin America and some Caribbean Islands
RIPE NCC	Europe, the Middle East, and Central Asia

www.iana.org

Classless Inter-Domain Routing (CIDR) Addressing Structure

- IP Address broken into two main parts
  - Network Part (net-id): ← External
    -
    -
  - Host Part (local part of the address) ← Internal
    - Subnet - network within a site or place
    - Host - host on the subnet



129.65.0.0/16

2^{16} hosts

Hierarchical Network Addressing
Longest Match (Longest Prefix Match)

Network Part of the address – CIDR Addressing

→ Classless InterDomain Routing

- **Network address**
  - **External address – how the rest of the world finds you**
  - Used for routing External to your local network
  - This part of the address gets it to your network

    - Who ever provides you with your addresses specifies the network part of the address

    - Cal Poly network part 129.65.0.0/16 → has 16 bits of freedom (32-16) how many bits in the network part
      - 16 bits in the external network part of the address

      - A packet address to 129.65.3.2 will arrive at Cal Poly
      - And a packet address to 129.65.87.3 will arrive at Cal Poly

/8 is the smallest
24 bits locally
↓
huge

CIDR Addressing – How to identify the network part of the address

- **How to identify the network part of the address**
  - The slash notation is called the **netmask**
    - So /16 is a netmask

  - **Netmask** says how many bits of the address are being used to route to the correct network.

    - Given IP: 129.65.10.5 with netmask: /16

      - First 16 bits of the address identify the network
      - Network part of address is: 129.65 (written as 129.65.0.0/16)

CIDR Addressing – Netmask examples

- Given IP: 77.195.10.20 with **netmask**: /10
  - What is the network part of this address

```
        77.       195.       10.       20
01001101.11000011.00001010.00010100     (IP)
11111111.11000000.00000000.00000000     (netmask of /10)
---------------------------------------------------------
77.       192.       0.       0
```

& 10 bl

So network is: 77.192.0.0 / 10

- Given IP: 44.89.130.89 with **netmask** /20
  - What is the network part of this address

44.89.130.89/20
& 255.255.1111 0000.00000000
44.89.128.0/20

Addressing Example

IP Address: 130.170.168.159 **Netmask**: /18

	130	.170	.168	.159
IP Address:	10000010.	10101010.	10101000.	10011111
Netmask (/18):	11111111.	11111111.	11000000.	00000000

Top tier Network-id (using the /18 netmask):

gets us to parent

Given a 2nd tier netmask of /22

- What is the 2nd tier network-id:

130.170.168.159 /22 = 130.170.168.0/
255.255.

- How many tier 2 networks are possible with this netmask?

22-18 = 4 = 2^4 = 16 children

Example continued on next slide

Addressing Example (con't)

IP Address: 130.170.168.159 Tier 2 **Netmask**: /22

130 .170 .168 .159

IP Address: 10000010.10101010.10101000.10011111

Netmask (/22): 11111111.11111111.11111100.00000000

Given a subnet mask of /25 (using a tier 2 ISP with netmask of /22)

- Decimal notation for this Mask: 255.255.255.128
- Subnet part of this address: 130.170.168.128/25
- Number of possible subnets: 25 - 22 = 3
- Host part of this address
- Number of possible hosts per subnet:

32 - 25 = 7 $2^7 - 2 =$ 126

Addressing Example #2

IP Address: 65.33.18.10 **Netmask**: /14

65 .33 .18 .10

IP Address: 01000001.00100001.00010010.00001010

Netmask (/14): 11111111.11111100.00000000.00000000

Top tier Network-id (using /14 netmask):

Given a 2nd tier netmask of /21

- What is the 2nd tier network-id:
- How many tier 2 networks are possible with this netmask?

Addressing Example #2 (cont.)

IP Address: 65.33.18.10 Tier 2 **Netmask**: /21

65 .33 .18 .10

IP Address: 01000001.00100001.00010010.00001010

Netmask (/21): 11111111.11111111.11111000.00000000

Given a subnet mask of /26 (using a tier 2 ISP with netmask of /21)

- Decimal notation for this Mask:
- Subnet part of this address:
- Number of possible subnets:
- Host part of this address
- Number of possible hosts per subnet:

CIDR Hierarchical Addressing Example – Address Aggregation

/8 → 255.0.0.0

/16 → 255.255.0.0

/19 → 255.255.192.0

/25 255.255.255.128

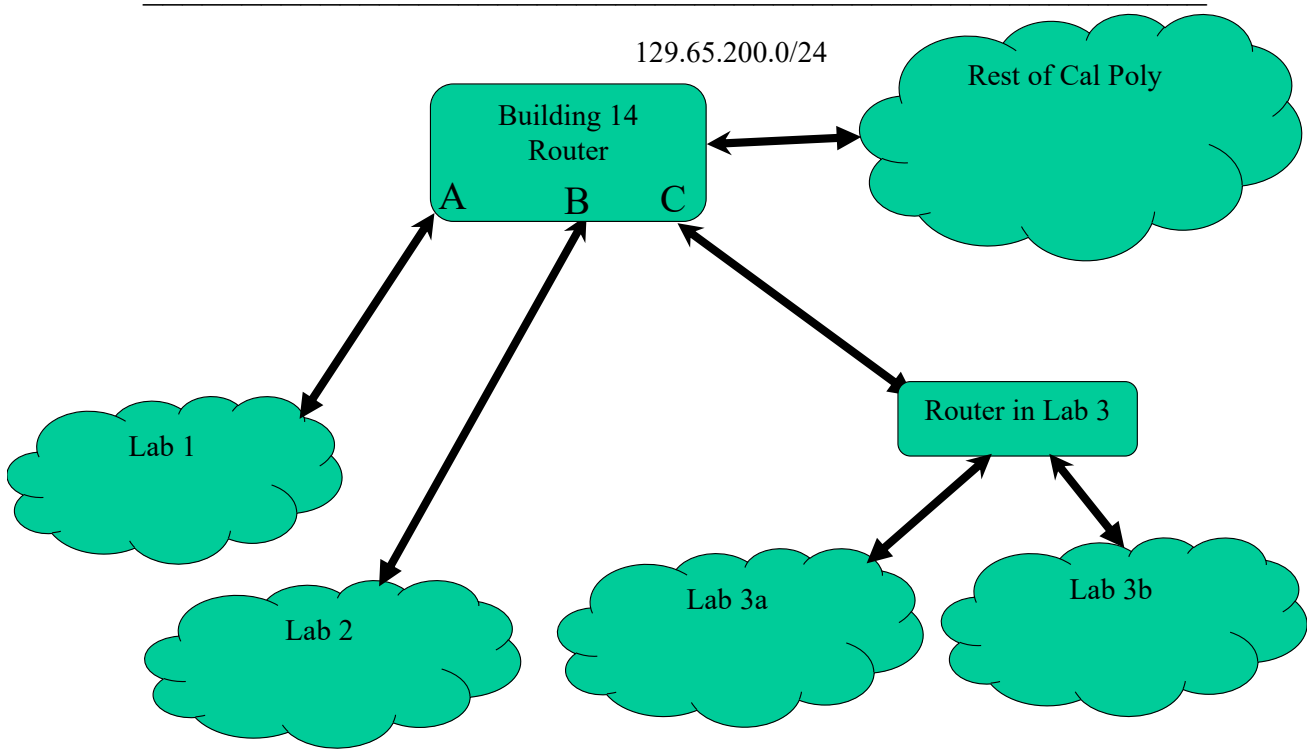
routers & PC use this notation

Example of Hierarchical Addressing

- As an example, ilets say that n building 14 we want four subnets
  - Subnet 1 – in lab 1
  - Subnet 2 – in lab 2
  - Subnet 3a – in lab 3
  - Subnet 3b – in lab 3

  - Building 14 address is: **129.65.200.0/24**
    - Assigned by ITS

CIDR Hierarchical Addressing Example – Address Aggregation



How to break up the addresses

- Starting with 129.65.200.0/24
  - Want to break this up into 4 subnets
  - Lab 1, Lab 2, Lab 3a, Lab 3B
  - How many bits do we need to have 4 subnets: _____

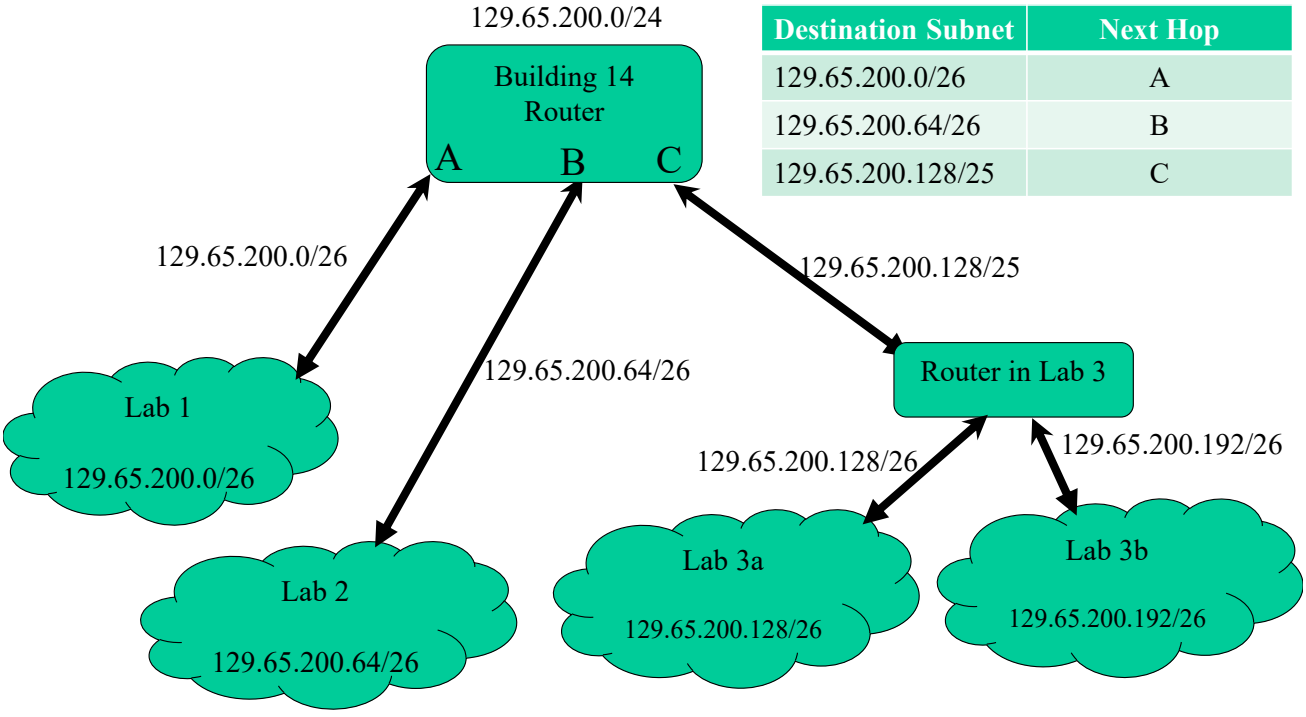
- 129.65.200.00000000 = 129.65.200.0/26

- 129.65.200.01000000 = 129.65.200.64/26

- 129.65.200.10000000 = 129.65.200.128/26

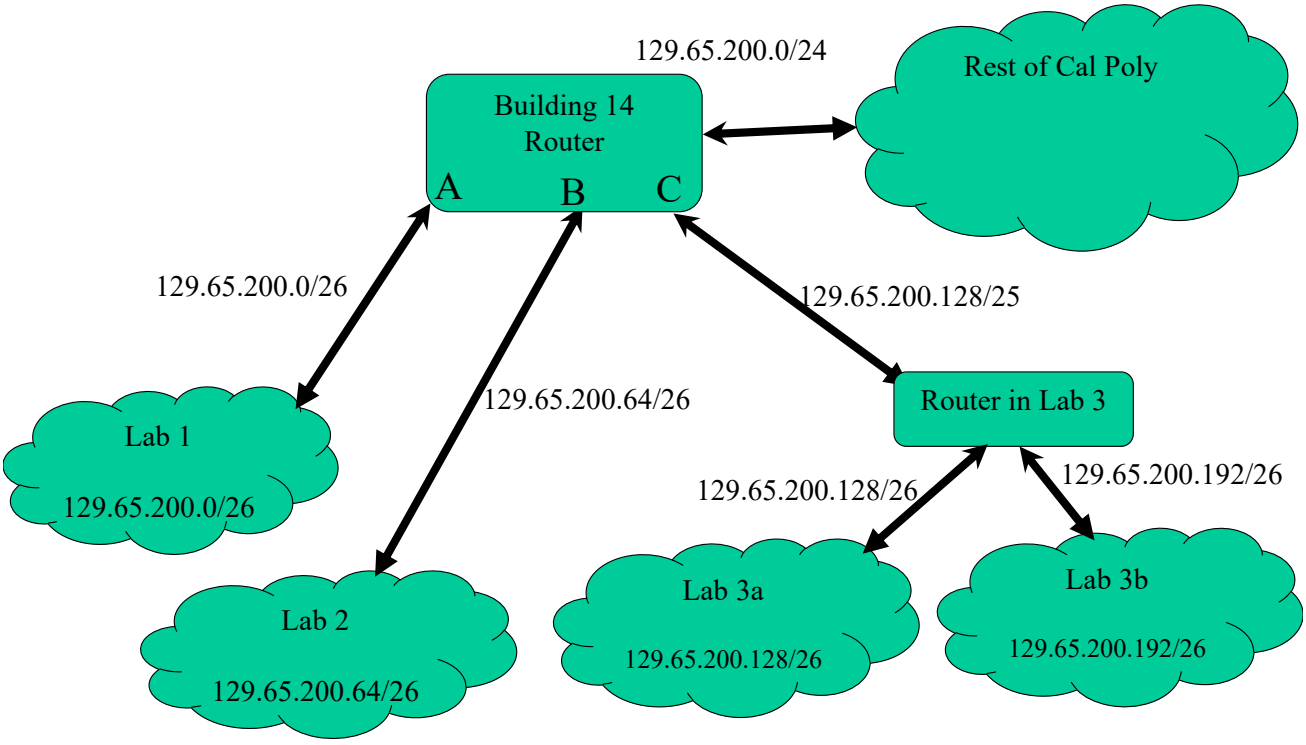
- 129.65.200.11000000 = 129.65.200.192/26

CIDR Hierarchical Addressing Example – Address Aggregation



Destination Subnet	Next Hop
129.65.200.0/26	A
129.65.200.64/26	B
129.65.200.128/25	C

CIDR Hierarchical Addressing Example – Address Aggregation



Some 129.6x addresses

IP range details

129.65.0.0/16

AS7960 · California Polytechnic State University

Summary

Country	United States
Domain	calpoly.edu
ASN	AS7960
Registry	arin
Hosted IPs	65,536
ID	CALPOLY

WHOIS Details

```
NetHandle:      NET-129-65-0-0-1
OrgID:          CPSU-1-Z
Parent:         NET-129-0-0-0-0
NetName:        CALPOLY
NetRange:       129.65.0.0 - 129.65.255.255
NetType:        allocation
RegDate:        1987-09-04
Updated:        2024-01-15
AbuseHandle:    NETWO381-ARIN
TechHandle:     IPADD11-ARIN
Source:         ARIN
```

https://ipinfo.io/AS7960/129.65.0.0/16

IP range details

129.66.0.0/16

AS3464 · Alabama Supercomputer Authority

Summary

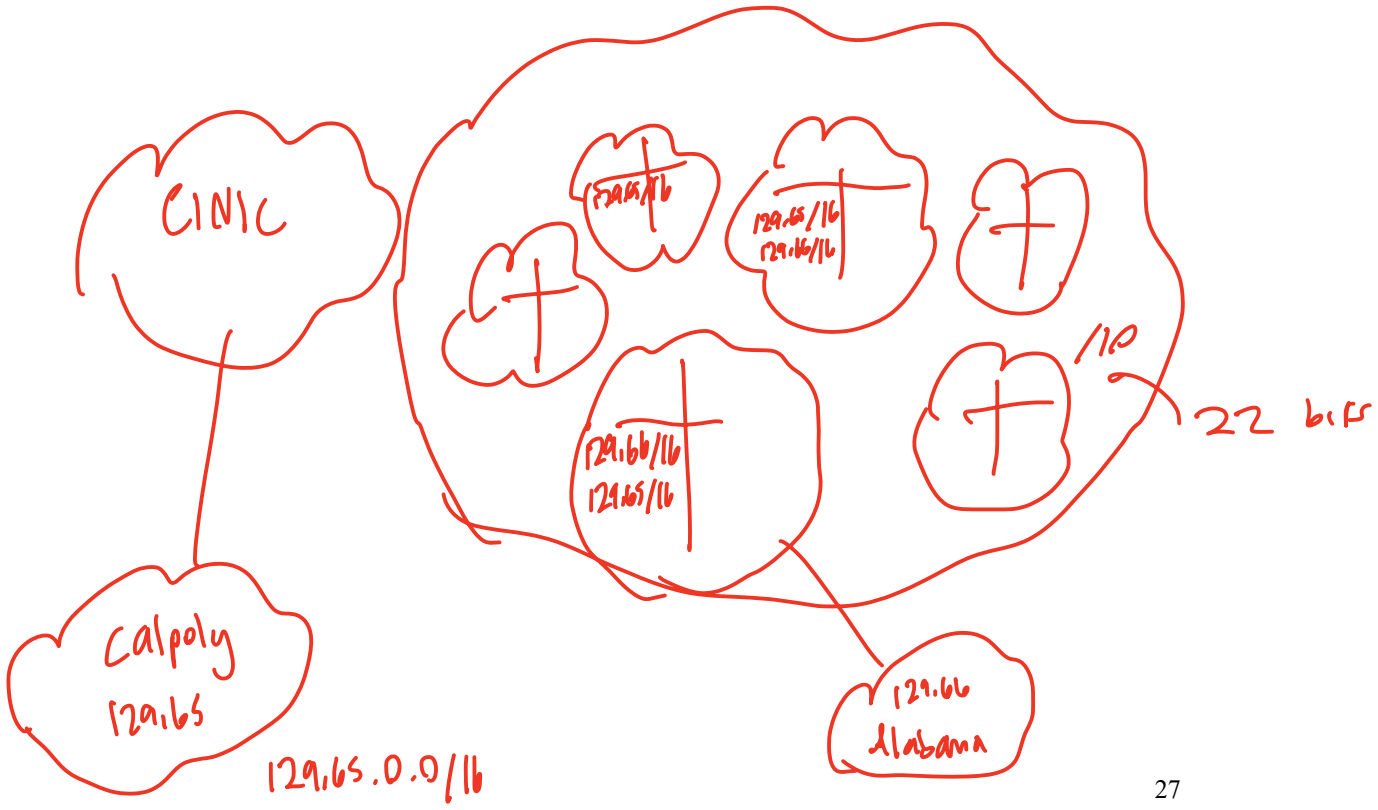
Country	United States
Domain	asc.edu
ASN	AS3464
Registry	arin
Hosted IPs	65,536
ID	ASC-UNIV-ALABAMA

WHOIS Details

```
NetHandle:      NET-129-66-0-0-1
OrgID:          ASC-Z
Parent:         NET-129-0-0-0-0
NetName:        ASC-UNIV-ALABAMA
NetRange:       129.66.0.0 - 129.66.255.255
NetType:        allocation
RegDate:        1987-09-05
Updated:        2023-11-30
TechHandle:     MD3-ORG-ARIN
Source:         ARIN
```

https://ipinfo.io/AS3464/129.66.0.0/16

How the World Routes to Cal Poly (how it really works)



Helps to break up routing

Site Local Addresses

- Addresses for private networks (Site local)
  - RFC 1918 provides for addresses that are only usable/routable locally
  - 10.0.0.0/8
  - 192.168.0.0/16
  - 172.16.0.0 – 172.31.0.0/12
  - Note: 127.0.0.1 is localhost (::1 is localhost in IPv6)
    - E.g. cclient myHandle **localhost** 55555
    - E.g. cclient myHandle **127.0.0.1** 55555

Classfull IP Addresses (older)

- Classfull IPv4 Addressing (older addressing scheme)
  - Still broke up address into two main parts
    - Net-id:
    - Host-id:
  - Differences:
    - Netmask was fixed based on class
    - Class of address was determined by the first few bits of the address
    - What we use today:
      - Class A uses a /8 netmask
      - Class B uses a /16 netmask
      - Class C uses a /24 netmask (but there are no real class C networks)

Class A

0	7 bits for network	24 bits for host

Class B

10	14 bits for network	16 bits for host

Class C

110	21 bits for network	8 bits for host

Classfull Addresses using CIDR

- Network addresses
  - Many address ranges in class A and Class B have not been taken (have not been reclaimed) for CIDR aggregate addressing
  - Class A:
    - Equivalent Classless network mask : /8
    - Some of the class A addresses have been reclaimed
    - CIDR allows the reclaimed addresses to be used by others
  - Class B:
    - Equivalent Classless network mask: /16
  - Class C:
    - Equivalent Classless Network mask: / 24
    - Class C addresses have been reclaimed and redistributed using CIDR aggregation addressing
  - In the core – all addresses are routed using CIDR longest match algorithm
    - Reclaimed addresses will have different masks
    - Un reclaimed addresses (traditional) will have /8 (class A) or /16 (class B)

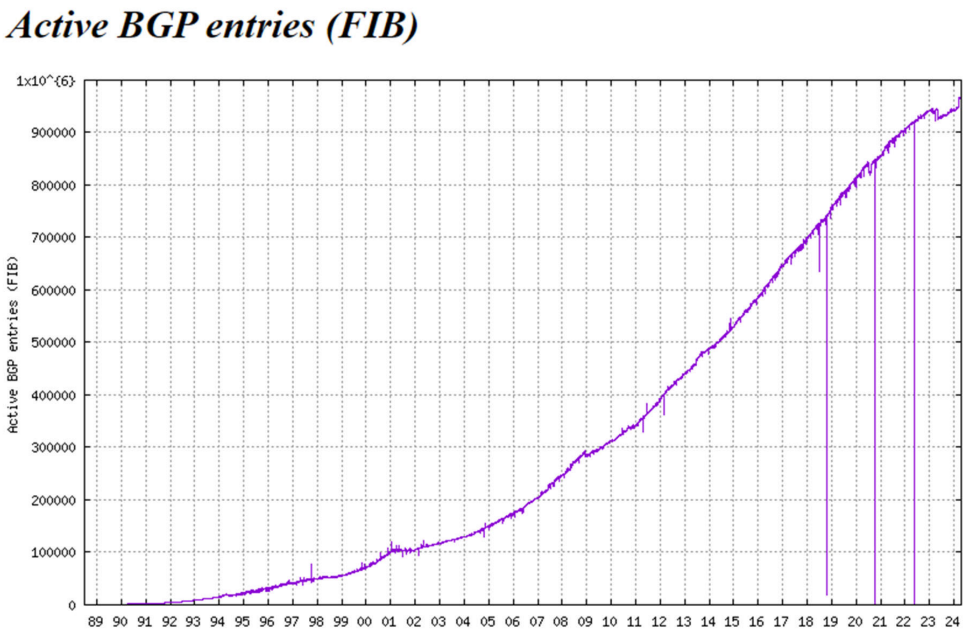
CIDR vs Classfull Addressing

- Problems with classfull addressing:
  - Network allocation sizes where fixed
  - The Internet was running out of IP addresses
    - Many wasted on larger class A and B networks
    - Class C was to small
  - The core routing tables where getting to large

- CIDR
  - Allows for a hierarchical addressing scheme – small core routing tables
  - Allows for variable sized networks
  - As part of CIDR reclaimed some class A's and all of class C
  - Allows people to break up their network space and sell it to others

- But CIDR is more complex to route
  - Classfull addressing can use a hash function to find routing table entry
  - CIDR requires a search of the table for the longest matching entry

Routes Shared by IPv4



BGP Forwarding Table (FIB)
(FIB=Forward Information Base)
http://bgp.potaroo.net/as2.0/bgp-active.html

CIDR – Longest Match Algorithm

Lookup: 88.168.101.33

88. 168. 103. 33
01011000.101010000.01100111.00100001

Which route to use?

IP Destination	NM	Next Hop Router
88.168.100.0 01011000.101010000.01100100.00000000	/24	16.34.72.1
88.168.103.0 01011000.101010000.01100111.00000000	/24	52.73.81.238
88.168.0.0 01011000.101010000.00000000.00000000	/16	175.32.126.93
88.168.0.0 01011000.101010000.00000000.00000000	/18	1.1.1.1
88.128.0.0 01011000.10000000.00000000.00000000	/9	209.23.7.1